

Drone swarm sensitivity estimation using Bayesian Theory for search and rescue operations

Luciano Lima¹, Ali Ghaderi², Fabio Andrade¹, Carlos Pfeiffer², Youcef Djenouri^{1,3}, Marcos Moura¹

¹University of South-Eastern Norway, Vestfold, Norway

²University of South-Eastern Norway, Porsgrunn, Norway

³Norwegian Research Center NORCE, Oslo, Norway

{limaluciano, fabio}@ieee.org, {ali.ghaderi, carlos.pfeiffer, youcef.djenouri, marcos.moura}@usn.no

Abstract—Search and rescue operations demand a fast and reliable process for locating victims. The use of unmanned aerial vehicles (UAVs) has become a prominent research topic due to their lower cost and operational flexibility compared to manned aircraft. This study investigates the use of Bayesian probabilistic methods to estimate the sensitivity of a drone swarm in maritime search and rescue missions. An initial sensitivity distribution is defined based on both prior exact and approximate knowledge, and is subsequently updated using observational data: the number of drones (out of 10) that successfully identified a survivor while flying over the same area. The model is iteratively updated and evaluated. Results show that even after just four missions, there is a consistent convergence of the mean and a reduction in the standard deviation from 0.1307 to 0.0449. These findings demonstrate a reusable framework that improves swarm sensitivity estimation even with limited data. The full code is available at <https://github.com/lima0luciano/bayesian-drone-sar>.

Index Terms—Hierarchical Bayes Model, drones, UAV, SAR operations

I. INTRODUCTION

One of the biggest challenging stages on a maritime accident is to locate the survivors [1]. The large area of an ocean demands the mobilization of a lot of human resources, vehicles and equipment reflecting in a considerable cost. The deployment of a U.S.Coast Guard response boat for performing an average pattern costs around \$5,000 while the use of an helicopter costs \$33,000 [2]. The use of

Unmanned Aerial Vehicles (UAV) with cameras is object of study of different researches currently [3]–[5]. By the use of computer vision algorithms, a swarm of UAVs can survey a large area with a lower demand of people and investment than the flight of airplanes and helicopters [6]–[8]. During a Search and Rescue (SAR) operation, the probability of the survivors location is set by the rescue team [1], [9]–[11]. Some of them use softwares which consider Bayes Theorem together with the environmental factors [12]. For any detecting activity an essential information is the capacity of the sensor measuring successfully the target information. On this work the sensitivity of a swarm will be explored using Bayesian theory. The interdependence of different systems in a UAV adds a lot of uncertainties in different levels. Hierarchical Bayesian Model is an strategy to approach complex problems in statistics written in hierarchical forms using Bayesian methods [13], [14]. It allows a better conceptual understanding and software design [15], offering a structured approach to iteratively evaluate system performance in real environments—an area of increasing relevance for detection systems development teams [16]. Furthermore, the statistical reproducibility provided by hierarchical Bayesian frameworks gives a robust and efficient foundation for accelerating algorithm development and performance evaluation [17].

Using a UAV swarm in SAR allows the optimization of resource usage, demanding the flight of helicopters only when the location of survivors is known [18]–[20]. Replacing manned aircraft by a higher number of drones allows to cover a wider area [8], [21]. In this context, a main question arises: What is the real performance of the proposed system for detecting survivors?

Applying a multi-agent system for detection faces difficulties in acquiring one specific value that represents its quality. Due to the innumerable uncontrollable factors, such as the position of the sun, wave heights, air visibility, among others, the scenario reflects a stochastic problem. The guarantee that a survivor will be identified is highly uncertain, but at the same time crucial for such a system. Despite the complexity of the multi-factorial problem, in this theoretical approach, a simple set of conditions was initially fixed to focus on the development of a Bayesian framework that is reproducible for application across different scenarios. This allows the creation of a theoretical foundation in a workable context to estimate the system's sensitivity. This background will allow further use with parameter variations in a more dynamic environment.

The factors that can influence a vision recognition system can be divided into three categories: environmental, hardware, and operational [22]–[25]. Environmental factors indicate uncontrollable conditions dependent of the scenario as lighting conditions, weather, air visibility and sea state. Hardware factors defined by the equipment capabilities have a direct impact on the success of detection like camera specifications, computer processing power, gimbal systems and battery capacity. Operational factors vary according to the mission coordination like the altitude, communication systems range and stability.

The aim of this work is to explain established Bayesian methodology in a controlled and simplified environment. Further research can be developed in the future to incorporate more complexities. The framework validation can be performed with simulated data, as it offers greater control and understanding of the underlying information, since

the data is generated by the research team. This makes it more suitable for evaluation purposes than using real-world data.

This work makes the following key contributions to the field of drone-based SAR operations:

- 1) We provide a reproducible Bayesian framework aimed to estimate the sensitivity (i.e., detection capability) of a drone swarm focused in SAR operation.
- 2) We introduce a Hierarchical Bayes Model incorporating the uncertainties generated from environmental conditions and sensor performance. This model considers the sensitivity as a Beta distribution with its parameters modeled as random variables.
- 3) We consider the progressive input of new data to improve the sensitivity estimation. Although the continuous update of the posterior probability, the uncertainties are reduced on the estimation, leading to a better prediction model.
- 4) We demonstrate the proposed framework by the use of simulated data, showing a narrowing in the posterior distribution as the model is exposed to new data. The high flexibility of the simulated data provides a powerful context in which new scenarios can be created and evaluated.
- 5) We create the background for further studies integrating additional factors as drone intercommunication, search coordination path-planning, dynamic environmental changes etc with the aim to provide a comprehensible model for SAR operations.

II. RELATED WORK

Bayesian theory has been central to SAR since its notable application during the search for the USS Scorpion submarine in 1968, where prior maps were generated using Monte Carlo methods to refine the search space [26]. This foundational work inspired contemporary systems like SAROPS (Search and Rescue Optimal Planning System), which is widely used by the U.S. Coast Guard. SAROPS integrates environmental data and incident reports to generate Bayesian-based search plans [27].

Later works showed that Bayesian methods are reliable on SAR operations [28] and modern re-

search expands this approach to apply it in drone swarms. Giacomossi et al. [29] demonstrated cooperative strategies among UAVs for SAR tasks, showing that distributed intelligence and coordination can improve coverage and detection efficiency in forest and terrain-based rescue scenarios. This complements earlier findings by Furukawa et al. [30], which emphasized the superiority of multiple autonomous vehicles over single-agent systems in recursive Bayesian search-and-tracking tasks.

Other works have focused specifically on UAV object detection performance in maritime conditions. Zhao et al. [22] introduced YOLOv7-SEA, a computer vision model adapted for UAVs operating over water surfaces, which is resilient for brightness and low-contrast conditions in maritime images, while Rakotondraibe et al. [23] developed an integrated computer vision and drone control system for drowning recognition under ocean surveillance scenarios.

SAR operations has several stochastic challenges such as variable lighting, sea states, and occlusions, that requires a robust probabilistic models. Bayesian frameworks, especially Hierarchical Bayesian Models, provide a compelling structure for such contexts. Veenman et al. [14] emphasized the usefulness of this approach in behavioral sciences and complex environmental modeling, showing their capability to integrate uncertainty at multiple levels.

Recent studies also explored sensitivity estimation in vision-based UAV systems. Bai et al. [25] applied improved YOLOv5 models to marine casualty detection, offering insights on how different configurations affect detection sensitivity. Shetty et al. [24] implemented multi-sensor drone strategies to improve detection accuracy, reinforcing the need for models that can dynamically adjust to variable operational performance.

The importance of modeling and continuously updating system sensitivity has been reflected in new Bayesian approaches that allow iterative refinement through posterior updates. This allows for performance adaptation as new data becomes available, an approach central to the present work. Similar adaptive Bayesian updating methods have been applied in marine UAV deployments for mapping

and tracking purposes, such as those demonstrated by O'Kelly [12] in spatial search contexts.

Despite advances, most work still treats UAV performance metrics statically. This study presents a Hierarchical Bayesian Model that captures sensitivity as a Beta-distributed variable, with parameters themselves modeled as distributions, allowing a dynamic and data-driven refinement process. The objective is to estimate and analyze the sensitivity of a drone swarm considering the presence of some uncontrollable external parameters that leads to a stochastic nature. This is essential for assessing the robustness and reliability of swarm control systems operating under real-world conditions such as environmental noise, communication delays, and hardware imperfections.

III. METHODOLOGY

A. Principle

The sensitivity improvement process is based on Bayesian updating represented in the Figure 1. Initially, a probability distribution function representing the swarm sensitivity is considered as a prior distribution. One random simulation is chosen from the set of simulations that represent how many drones from the swarm were successful in finding the survivor. Through the Bayesian updating process, a new swarm sensitivity is calculated as a posterior distribution, reflecting a more accurate function that describes the performance of the system. The quality of the sensitivity estimate is improved by incorporating the new information available from the simulation. The process can therefore be repeated, consecutively improving the system evaluation.

This is a toy problem designed to reflect the uncertainties of the reality once a long range of factors can influence the detection success on a real world scenario. Considering the stochastic nature of this problem, the Bayes Theorem will be used to address the uncertainties of the survivor detection with the swarm of drones. A prior distribution of the swarm sensitivity will be updated based on the data of N missions performed by the swarm in a simulated environment, providing a posterior sensitivity distribution. Subsequently, with the availability of

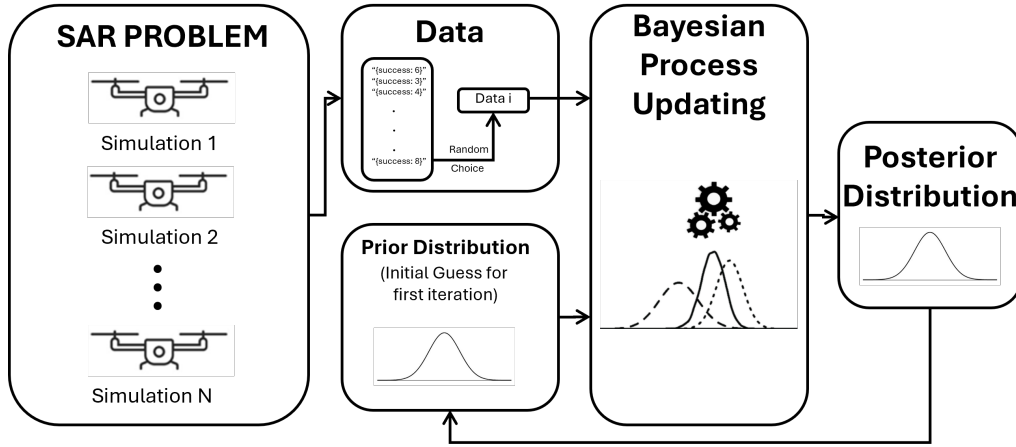


Fig. 1: Sensitivity improvement process

a new data, the process can be repeated aiming to reach to a narrower sensitivity estimation. The data represent the number of drones that found the target, represented by the structure “{success: s ””, where $s \leq N$. The process can be seen on the diagram in Figure 1.

B. Statement of Inference

After completing a mission, we get two key numbers: n , the total number of drones participating in the mission, and s , the number of drones successfully detecting the survivor. Based on these values, we aim to estimate the detection success rate, or swarm sensitivity, denoted by q , which lies between 0 and 1. The probabilistic formulation of the problem is given by:

$$P(q | s, n) \quad (1)$$

Which represents our degrees of belief about the possible values of q , given the observed n and s . For this discussion, it is assumed that the reader is familiar with standard probability notations and concepts. For further details, refer to [31]. Using the Bayes’ rule, we have:

$$\begin{aligned} p(q | s, n) &= \frac{1}{Z} p(s | q, n) \times p(q) \\ &= \frac{1}{Z} L(q; s, n) \times p(q) \end{aligned} \quad (2)$$

where, $L(q; s, n)$ is the likelihood, $p(q)$ is the prior distribution and Z is the normalization constant.

C. Likelihood selection

In general, the likelihood represents the data-generation process. In this case, the likelihood corresponds to the sampling distribution, which answers the following question:

“What is the probability that s drones successfully detecting the survivor, given q and $n \geq s$?”

As previously mentioned, the functional form of the likelihood aligns with the sampling distribution but is expressed as a function of q rather than s . To derive this, note that the probability of a single drone detecting the survivor is q , and this event is independent of all other drones. Consequently, the probability of s drones detecting the survivor while $(n - s)$ do not is given by:

$$p(s | q, n) \propto q^s (1 - q)^{(n-s)} \quad (3)$$

D. Prior selection

The prior is a feature of Bayesian inference, referring to the knowledge or beliefs held before observing new data. In this case, the prior addresses the following question:

“If we had previously observed certain number of drones under similar conditions and the corresponding number of successful detections, what could we say about the value of q before seeing the new observations?”

The answer depends on the experience of the investigator and is, therefore, subjective. Nevertheless, it can be argued that the distribution representing the degrees of belief for this question should belong to a specific family of distributions, as induced by the data-generating model. Following the method of likelihood-induced priors [32], can be written that:

$$p(q) = \frac{1}{B(\alpha, \beta)} q^{\alpha-1} (1-q)^{\beta-1}, \quad (4)$$

for some $\alpha, \beta > 0$

where $B(x, y)$ is the beta function. This implies that the prior distribution of q belongs to the Beta family. In this context, the investigator prior belief about q is modeled by specifying values of α and β . To this end, note that:

$$p(q) \propto q^{\alpha-1} (1-q)^{\beta-1} \quad (5)$$

Using the same approach of defining the sampling distribution, $(\alpha - 1)$ and $(\beta - 1)$ can be interpreted as the quantity of drones that performed successful and failed detections of the survivor, respectively. It will be assumed that the values of α and β are known once these values are part of the dataset.

1) *Posterior distribution:* Building on the above, the posterior distribution after observing one mission is given by:

$$p(q | s, n) = \frac{1}{Z} L(q; s, n) \times p(q) \quad (6)$$

$$= \frac{1}{Z} q^{(s+\alpha-1)} (1-q)^{(n-s+\beta-1)}$$

Upon calculating the normalization constant, the posterior distribution simplifies to:

$$p(q | s, n) = \frac{1}{B(\alpha + s, \beta + n - s)} q^{(s+\alpha-1)} \times (1-q)^{(n-s+\beta-1)} \quad (7)$$

IV. RESULTS

To analyze the proposed solution, simulated data will be used, generated from a random process with unknown probability of success to detect the survivor. The data is given in Table I.

The scenario considered on this work is based on the flight of a swarm of 10 drones aiming to find one survivor on the open sea. All drones will fly over the same area at same altitude performing different random paths and without any kind of communication between them. The data used on calculations is composed by a sequence of numbers representing successful flights performed by the drones on the fleet. A flight of a specific drone is considered a success when it finds the survivor. Different environmental factors can influence the detections. There were done 100 flights of the fleet, the amount of drones which detects the survivor is mapped as the sensitivity of the swarm.

The data related to 3 missions will be used for this study aiming to demonstrate how beliefs evolve with an increasing number of missions. The small number of missions is used due to the consideration of the difficulties for performing them in real world, so the main idea is to see the impact of just a few of new data. Three random data points were acquired from the original file representing the missions 26, 31, and 68

Mission number	Number of Successful drones (s_i)	Number of Failed drones (f_i)
1	7	3
2	6	4
3	9	1

TABLE I: Performance of drones in different missions.

A. Prior based on a priory exact prior knowledge

Now, let us assume that, based on previous observations or theoretical considerations, we believe that 7 out of 10 drones will successfully detect the

survivor. That is, $\alpha = 8$ and $\beta = 4$. Under this assumption, according to the Equation 4 the prior distribution is given by:

$$P_0 = p(q) = \frac{1}{B(8, 4)} q^7 (1 - q)^3 \quad (8)$$

and hence, the posterior distribution after one mission, representing our updated belief, is:

$$P_1 = p(q | s_1, n_1) = \frac{1}{B(8 + s_1, 4 + n_1 - s_1)} \times q^{(s_1+7)} (1 - q)^{(n_1-s_1+3)} \quad (9)$$

For the first mission, we observe that 7 out of 10 drones successfully detect survivor. This implies that the posterior distribution after the first mission is:

$$P_1 = p(q | s_1 = 7, n_1 = 10) = \frac{1}{B(15, 7)} q^{14} (1 - q)^6 \quad (10)$$

Figure 2 demonstrates the effect of this observation on our belief after one mission. Notably, the posterior distribution is narrower than the prior distribution, indicating that the uncertainty about the true value of the q is decreased after the first mission.

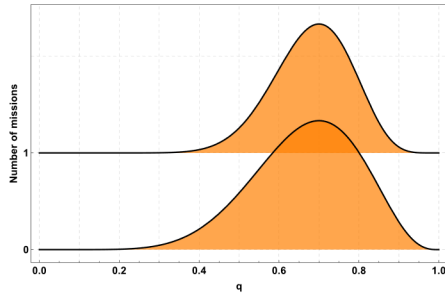


Fig. 2: Plot of the prior (zero mission) and the posterior (after one mission) distributions of the q

Next, additional missions are conducted, and we collect more data. After each mission, our belief about q is updated. Using the Bayes' rule, the posterior distribution after the second mission (represented by s_2, n_2) becomes:

$$P_2 = p(q | s_1, n_1, s_2, n_2) = \frac{1}{Z_2} p(s_2 | q, s_1, n_1, n_2) \times p(q | s_1, n_1, n_2) \quad (11)$$

Since the information about n_2 is redundant, the new prior becomes

$$p(q | s_1, n_1, n_2) = p(q | s_1, n_1) = P_1 \quad (12)$$

which is the posterior distribution after the first mission. Furthermore, since the missions are independent, we have:

$$p(s_2 | q, s_1, n_1, n_2) = p(s_2 | q, n_2) = L(q; s_2, n_2) \quad (13)$$

where $L(q; s_2, n_2)$ denotes the likelihood. Hence:

$$P_2 = \frac{1}{Z_2} L(q; s_2, n_2) \times P_1 \quad (14)$$

Following the same logic as for the first mission, we obtain:

$$P_2 = \frac{1}{B(\alpha + s_1 + s_2, \beta + (n_1 - s_1) + (n_2 - s_2))} \times q^{(s_1+s_2)+\alpha-1} \times (1 - q)^{(n_1-s_1)+(n_2-s_2)+\beta-1} \quad (15)$$

Similarly, after k missions, the posterior distribution is given by:

$$P_k = \frac{1}{B(\alpha + S, \beta + F)} q^{(S+\alpha-1)} (1 - q)^{(F+\beta-1)} \quad (16)$$

where:

$$S = \sum_{i=1}^k s_i \quad \text{and} \quad F = \sum_{i=1}^k (n_i - s_i) \quad (17)$$

represent the total number of success and failures across k missions, respectively. Since $n_i \geq s_i$, as the number of missions increases, the influence of α and β diminishes. Indeed, note that:

$$k \rightarrow \infty \Rightarrow \alpha + S \rightarrow S \quad \text{and} \quad \beta + F \rightarrow F \quad (18)$$

This means that as the number of missions grows, the posterior distribution becomes increasingly dominated by the data—a hallmark of a good learning process. With each new piece of evidence, our belief is updated accordingly. Figure 3 shows the evolution of the posterior distribution for q as the number of missions increases from 0 to 3, using $\alpha = 8$, $\beta = 4$ and the data provided in Table I.

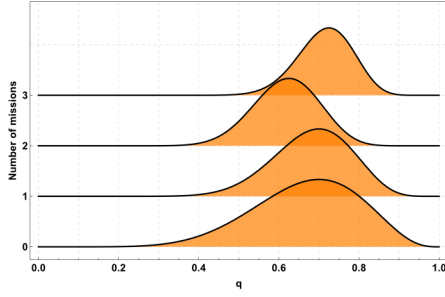


Fig. 3: Plot of the prior (zero mission) and the evolution of the posterior distributions with increasing number of missions

B. Prior based on a priory inexact prior knowledge

In the previous section, we assumed that we had prior knowledge suggesting that 7 out of 10 drones would successfully detect the survivor. However, it is common to lack such precise knowledge. Instead, more often, the previous knowledge is imprecise. For example, assuming that the observations are made under similar circumstances does not necessarily imply identical outcomes for every mission. For instance, while weather conditions might be similar, they are rarely exactly the same, and therefore variations in outcomes can be expected. In general, we do not know the exact values of α and β but we have some idea about their possible ranges. Nevertheless, our primary interest lies in q , not α or β . However, knowledge about α and β is necessary to estimate q , which is why α and β are commonly referred to as nuisance parameters. Their values are not of primary interest, but this lack of precise knowledge generally leads to greater uncertainty in the inferred value of q . To this end, recall that, as mentioned previously, $(\alpha - 1)$ and $(\beta - 1)$ can be interpreted as the number of drones detecting and not detecting the survivor, respectively. Thus, in our case,

$$(\alpha - 1) + (\beta - 1) = 10 \Rightarrow \alpha + \beta = 12 \quad (19)$$

This means that α and β are linearly dependent. For this reason, we focus on α . Note that:

$$1 \leq \alpha \leq 11 \quad (20)$$

In the following, we relax the condition that α must be an integer. In reality, available information is often expressed as averages within a range of possible variations, which are not necessarily integers. To this end, we propose the following linear model:

$$\begin{aligned} \alpha &= 10r + 1 \\ r &\sim \text{Beta}(16, 7) \end{aligned} \quad (21)$$

This implies that the expected value of α is 7.96 with the 90% credible interval being [6.32, 9.15]. In other words, theoretically α can have any value between 1 and 11. However, the model assigns the highest probability to values within the range of 6 to 9. The probable values of α , and hence β , will be inferred in the analysis in order to estimate q . This approach is a standard example of Bayesian hierarchical modeling. The current problem is modeled with Bayesian statistics for estimating the value of q , which is described by a Beta distribution, which in turn is composed by parameters also described by probability distributions. Describing this problem in an hierarchical structure can be represented as shown below:

From a Bayesian point of view, if the exact value of a parameter is unknown, inference should be conducted by averaging over all possible values of the parameter. This principle is formalized as the marginalization rule [31]. In the present case, we have:

$$\begin{aligned} p(q | S, F) &= \int_1^{11} \int_1^{11} p(q, \alpha, \beta | S, F) d\beta d\alpha \\ &= \int_1^{11} \int_1^{11} p(q | \alpha, \beta, S, F) p(\beta | \alpha) \\ &\quad \times p(\alpha) d\beta d\alpha \\ &= \int_1^{11} P_k p(\beta | \alpha) p(\alpha) d\beta d\alpha \end{aligned} \quad (22)$$

Given that knowledge of α determines β , we have:

$$p(\beta | \alpha) = \delta(12 - \alpha) \quad (23)$$

where δ represents the delta function, denoting the absolute certainty.

Although this integral can be solved numerically for different values of q , we implement a Markov Chain Monte Carlo (MCMC) procedure known as Hamiltonian Monte Carlo (HMC) to estimate the posterior distribution of q . The programming platform Stan was used to generate the samples [33]. The Stan model requires the following input:

$$\begin{aligned} r &\sim \text{Beta}(16, 7), \\ \alpha &= 10r + 1, \\ \beta &= 12 - \alpha, \\ q &\sim \text{Beta}(S + \alpha, F + \beta). \end{aligned} \quad (24)$$

In Figure 4, the marginal posterior distribution of q is shown. For comparison, the posterior distributions of q are presented for both cases: exact and inexact priors. The distributions are similar, with the inexact prior yielding a slightly broader distribution. This confirms that the data sufficiently constrains the posterior such that the inference on q is not overly sensitive to the choice of α and β .

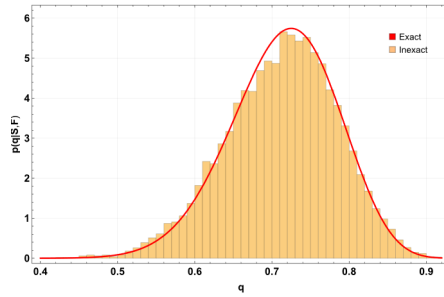


Fig. 4: The marginal posterior distribution of q with exact and inexact prior information

V. DISCUSSION

A. Model Validation

As shown earlier, the posterior distribution is updated as more information becomes available. However, even after three missions, a considerable amount of uncertainty remains about q . This uncertainty arises from at least two factors. The first one is due to limited data. The dataset contains only three missions. Generally, more data would result in narrower posterior distribution. The second factor is the complexity of q . In reality, q depends

on various environmental and instrumental factors. Estimating q therefore relies on knowledge of these factors. In the absence of such knowledge, assuming a single true value for q is no longer appropriate. Consequently, this assumption impacts both the estimate of q and the associated uncertainty. A more holistic approach would model these factors and their influence on q , but this is beyond the scope of this article and will be addressed in future work. Nonetheless, any estimate of q should be validated against new measurements. Specifically, we ask the following question: “If 10 drones are deployed under environmental conditions similar to those of the previous three missions, what is the probability distribution for the number of drones successfully detecting the survivor, given the results of the first three missions?”

This question can be addressed using the basic product and sum rules of probability theory. Indeed, we aim to describe $p(s_{\text{new}} | S, F)$, where s_{new} is the number of successful detections in the new mission, and S and F are the cumulative success and failures from the previous missions. Using the product rule and the marginalization:

$$\begin{aligned} p(s_{\text{new}} | S, F) &= \int_0^1 p(s_{\text{new}}, q | S, F) dq \\ &= \int_0^1 p(s_{\text{new}} | q, S, F) p(q | S, F) dq \quad (25) \\ &= \int_0^1 p(s_{\text{new}} | q) P_3 dq. \end{aligned}$$

where $s_{\text{new}} \sim \text{Bin}(10, q)$ and P_3 is the posterior distribution of q after three missions (as given in Eq. 16). Note that $p(s_{\text{new}} | q, S, F) = p(s_{\text{new}} | q)$, as knowledge of q makes prior observations redundant. Calculating this integral yields:

$$p(s_{\text{new}} | S, F) = \binom{10}{s_{\text{new}}} \times \frac{B(s_{\text{new}} + S + \alpha, 10 - s_{\text{new}} + F + \beta)}{B(\alpha + S, \beta + F)} \quad (26)$$

Using the data from the first three missions and the assumed values of $\alpha = 8$ and $\beta = 4$, this becomes:

$$p(s_{\text{new}} | S, F) = \binom{10}{s_{\text{new}}} \frac{B(s_{\text{new}} + 30, 22 - s_{\text{new}})}{B(30, 12)} \quad (27)$$

This distribution, also known as the posterior predictive distribution for s_{new} , is shown in Figure 5

and is denoted by $p(s_{\text{new}} | S, F)$. The plot illustrates that the uncertainty in q propagates to the predicted number of successful detections. Notably, there is a slightly greater than 47% chance of observing 7 or 8 successful detections. However, the probability of other outcomes is nearly 53%, resulting in a 90% credible interval of $[5, 9]$. This means it is quite likely that most future observations will fall within this range. For instance, in a subsequent mission where 10 drones were deployed under similar conditions, 9 drones successfully detected the survivor (red line in Figure 5). According to the model, the probability of 9 out of 10 drones succeeding is 15.3%. While this outcome is approximately 36% lower than the most likely value, it remains within the 90% credible interval. This highlights the importance of not relying solely on the most probable value of q . In planning and decision-making, it is crucial to consider the full distribution of q to account for the uncertainty surrounding its true value.

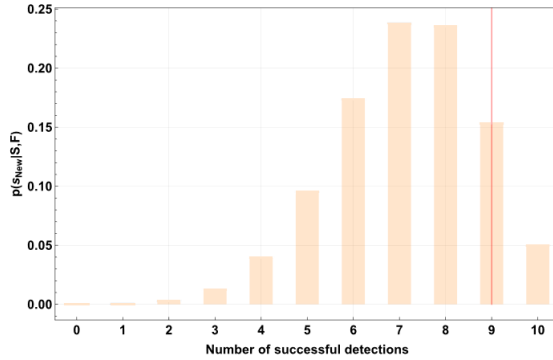


Fig. 5: Posterior predictive distribution for s_{new} with exact prior

Considering the values present in Table II, it is possible to see the mean gradually increasing from 0.6667 to 0.6944. The variation and the standard deviation show a consecutive decrease, dropping respectively from 0.0171 in Mission 0 to 0.0029 in Mission 3, and from 0.1307 to 0.0449. This suggests not only better performance over time but also greater certainty about the sensitivity in each new mission executed.

Mission	Mean	Variation	Standard Deviation
Mission 0	0.6667	0.0171	0.1307
Mission 1	0.6818	0.0094	0.0971
Mission 2	0.6667	0.0052	0.0719
Mission 3	0.6944	0.0029	0.0539

TABLE II: Mean, variation, and standard deviation for each mission.

VI. CONCLUSION

The addition of data to the model showed a real increase in certainty, allowing the model and the distribution to be optimized to better reflect the problem. The narrowing achieved for the sensitivity q in both cases of having exact and inexact prior knowledge provides a better guess for this value, allowing better modeling and further exploration of its optimization and simulations. The sensitivity progression can also serve to provide a good indication of the amount of observational data that will not be able to improve the model anymore, avoiding running simulations that do not lead to a better model. Further work can be done considering active communication between the drones, path coordination for the fleet, and environmental changes—variations in the factors currently considered uncontrollable.

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